

SECURITY PROBLEMS AND SOLUTIONS IN DATABASES**İsa AVCI**

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ORCID ID: <https://orcid.org/0000-0002-6048-7645>**Abstract**

Database security has become an issue that should be given importance in today's times when the need for data storage is at a high level. The need for data storage, which has developed with technology, has brought along attacks caused by database security vulnerabilities. As many new storage methods and protection methods have been developed depending on the developing technologies over time, new attack methods have emerged due to the security vulnerabilities found. Due to the lack of a conscious user and company structure, even many small cyber attacks can be successful because very simple precautions are not taken. One of the reasons why database security is not at the forefront is the lack of sufficient research and development studies in this area. This study, it is aimed to ensure and protect the security of the data stored in terms of the database. The concepts of integrity, confidentiality, and accessibility, which are the main elements of information security, are at the forefront of all security issues. Wide authorization, weak passwords, and patch deficiencies were identified as the most common security problems in databases. In addition, it is aimed to inform and raise awareness about security problems and solutions in databases. The most common security problems in databases were analyzed, and the solutions for the related problems were discussed. International security standards and secure models have been analyzed for cyber attacks against databases.

Keywords: Database security, cybersecurity, database cyber-attacks.**1. INTRODUCTION**

Databases are information repositories that need to be protected where sensitive information of institutions and individuals is stored. The need for database usage has increased day by day because data storage has become important with the developing technology. Today, all applications, from the simplest to the most complex, need data storage. With the increasing use of databases, cyber-attacks have also shown a high increase (Doherty et al., 2009; Ponemon, 2007). The inadequacy of the measures taken against these cyber attacks has revealed security problems (Aslay, 2017). At this stage, apart from the measures taken at the technical level, a conscious user infrastructure should also be created. It is necessary not to ignore the conscious user structure in these structures, where user-based problems and individual mistakes cause much material and moral damage.

There are common security problems that have been accepted as a result of research on database systems. Related problems are common on most systems and are a major factor in most attacks failing. These problems are; it can be listed as excessive authorization, weak password definitions, missing updates, auditing weaknesses, using default user names, excessive privileged stored procedures, lack of SSL structure, allowing unencrypted logins, and using similar passwords (Mousa et al., 2020).

In this study, information was given about the database and database security in general. Afterward, the security problems of database systems and the proposed solutions to these problems were mentioned, technical information was given about the security infrastructure and it was aimed to raise awareness on the human factor side of the problems. Explaining all these issues at a detailed level, will contribute to the adequate level of security measures regarding database security and will create awareness in terms of database security.

2. SECURITY ISSUES IN DATABASES

Databases are areas where data is stored and with the increase in the need for information storage, there has been a need for new methods of storing and accessing information in computer environments. It has emerged as an alternative to the traditional file-processing method. Phone book structure can be given as a simple database example (Imran and Hyder, 2009; Ramakrishman and Gehrke, 2003). It is a matter of protecting the confidentiality, integrity, and access of sensitive information stored in databases against attacks. The content of database security coverage can be detailed in various types and categories, such as technical and administrative. Database security is an issue that requires expertise and competence in many fields (Mousa et al., 2020).

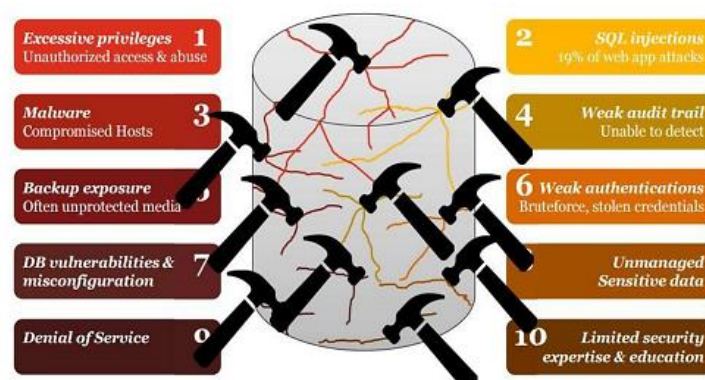


Figure 1. Database Threats (Mousa et al., 2020).

Database protection can provide secure and safe access to database information, in addition to ensuring data confidentiality, accuracy, and overall efficiency. Database security aims to ensure that only approved users execute authorized tasks at authorized times.

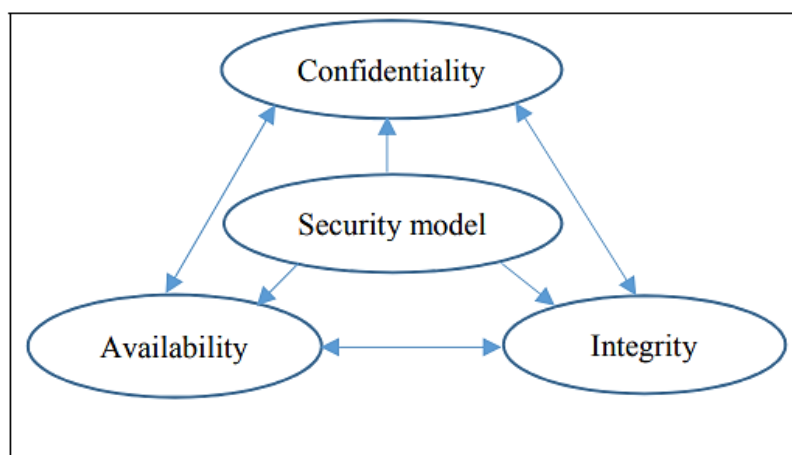


Figure 2. CIA Triad (Mousa et al., 2020).

Security issues in databases can be diversified in many ways. Today, different classifications have been made for these problems. However, the most common security problems with solutions; are the abuse of excessive privileges, risks arising from the database environment, and threats to data privacy (Vural and Sağiroğlu, 2007).

Authorizing users other than the authorizations entrusted to them brings great difficulties. In such cases, users can abuse their powers. In a university's database system, the fact that the IT officer has the authority to access and edit the data in the financial field means that he can change the payment information about the students. In financially focused environments such as banks, where more damage may be incurred, such authorizations can lead to greater problems. Having such excessive privileges does not necessitate abuse, but it can also create large gaps in the case of theft of a password that develops outside the user's administration.

The environments in which the databases are located are of great importance for the security of the database system. Security vulnerabilities in operating systems can cause loss, corruption, and DDoS attacks in databases (Bertino and R. Sandhu, 2003). For example, a vulnerability in the Windows 2000 operating system caused the Blaster worm malware to spread in the systems and caused intense DDoS attacks (Morrison, 2003).

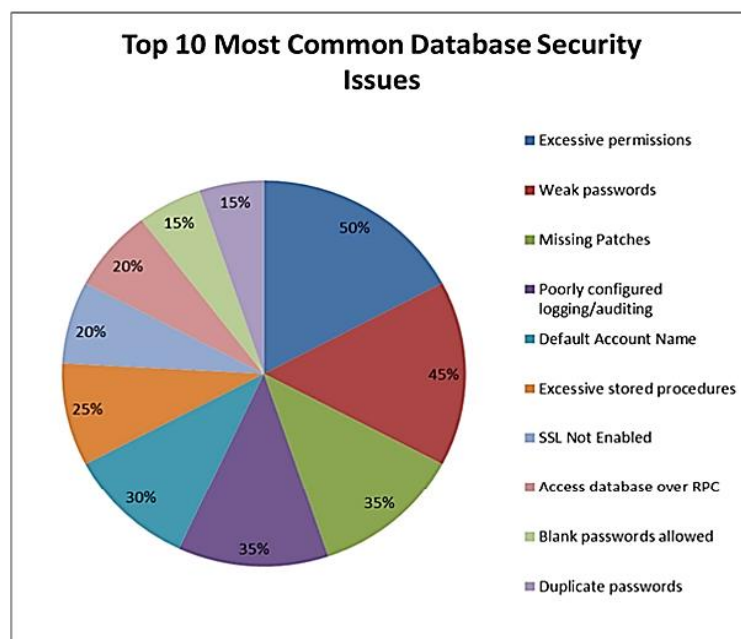


Figure 3. Top 10 most common database security issues (Malik and Patel, 2016).

Threats to data privacy consist of factors that prevent the protection of data privacy. It makes groups these threats under two main headings as SQL injection attacks and internal threats. Access to the database system is provided by adding special parameters to the SQL statements designed for validation in injection attacks. This attack is usually done using stored procedures. Many systems make queries by connecting to database systems in control mechanisms. If the information is verified as a result of the query, access is allowed. When the query structure is changed in abuse, access to the database is provided by obtaining validation approval from the system.

3. SECURITY MEASURES IN DATABASE

Security problems in databases have made it necessary to develop solutions to these problems. Today, many solution methods have been proposed for these problems, including artificial intelligence methods (Morrison, 2003). Solution proposals for security problems gathered under the heading of Problems are stated under the same headings.

There are access control mechanisms for security problems caused by excessive privileges. Access control mechanisms are methods that aim at the confidentiality of data. When a user wants to access any data, the relevant access control mechanisms are activated and control the rights of the relevant user. These rights are set by a database administrator or security administrator. A strong access control mechanism should authenticate users and prevent access by unauthorized users. Mechanisms; We can classify them as optional control mechanisms, role-based access controls, enforced control mechanisms, and mechanisms that notify attacks (Gupta and Agrawal, 2017).

Users can define authorizations for other users. Many institutions use this method because of the flexibility of the model. Authorizations can be defined and deleted by the administration. There are different mechanisms in the approach. It can list them as system R authorization model, content-based access control, and line-based access control.

System R Authorization Model: In this model, each user can create a new table and in the tables, it creates the relevant user is fully authorized. In this table, where he has full authority, he can define authorization for another user. In the default structure of the system, a user who is fully authorized in the table can also authorize another user to authorize an external user,

specific to the table. When the authorizations given here are withdrawn, the authorizations given to someone else by the person whose authorization is taken are taken back in the mirror with an authorization tree structure. However, as a result of the management of this method under the name of Non-cascading, when a user's authorization is taken, the authorizations given to others by the person whose authorization is taken are not taken (Bertino and Sandhu, 2005; Morrison, 2003; Gupta and Agrawal, 2017).

Content-Based Access Control: In this method, access control is provided according to the data content. Access to master tables is removed directly. Temporary tables fed from main tables are created by using the View structure instead of direct access. The content of the temporary table created according to the needs of the user is determined and the users access the database with the created View. Additional security is provided due to the closure of the direct access path to the main tables (Bertino and Sandhu, 2005; Morrison, 2003; Gupta and Agrawal, 2017).

Row-Based Access Control: This method has similar features to content-based access control. The model also creates temporary tables using the View structure and manages communication without access to the main tables according to the user's needs. Unlike the other model, it has the feature of fetching user-specific data. However, the operation of the model necessitates a related column that can be matched in the relevant main tables. View structures are prepared over the column to be associated, and the user provides access to information with filtering methods (Bertino and Sandhu, 2005; Morrison, 2003; Gupta and Agrawal, 2017).

Role-Based Access Controls: Users make it possible to create role templates on role-based model databases. These role structures can be assigned to users. The ability to assign new users to the determined patterns provides a great advantage in the first definitions. At the same time, it significantly reduces the workload of the role update to be made for a certain role group. Because it provides a certain order, all users in the system are assigned a role, so the access rights of the users are certain (Bertino and Sandhu, 2005; Morrison, 2003; Gupta and Agrawal, 2017).

Mandatory Control Mechanisms: This method is defined as controlling data and users by classifying them. All data is marked with confidential, top secret, and public labels. In this way, the order of importance of the data is created. Users are given the authority to read and write data marked according to the importance tag. If a user has access to the data with the top secret label, the criterion is sought whether he/she also has the right to read and write the data he/she wants to process. In order of importance, there is a hierarchy in the data. If a user has permission to read and write top-secret data, it means that he has permission to access all data in the system. Access to the highest level of information hierarchically provides this authority to the user. However, the same is not the case for the user who has permission to the marked data in general. It cannot access top-secret objects with a higher order of importance because they do not have read and write permissions (Bertino and Sandhu, 2005; Morrison, 2003; Gupta and Agrawal, 2017).

Mechanisms to Notify Attacks: In this approach, each user has an authorization definition in the system. Under normal circumstances, it is not noticed that users with the relevant authorizations try to hijack their passwords or try to abuse them. When a user tries to access an unauthorized area in the building, the system sends a warning to the responsible person with the warning mechanism. This warning structure can be in the form of an e-mail or short message. In this way, the relevant user may be responsible for the thought that he or she is inclined to abuse or that his password may have been compromised (Bertino and Sandhu, 2005; Morrison, 2003; Gupta and Agrawal, 2017).

Today, since every system has a certain level of vulnerability, operating system companies are trying to close these gaps with great effort. For this reason, studies with new patches and versions are published moment by moment. The emergence of a vulnerability due to any attack on a system means that all administrations with this system may face the same threat. For this reason, it is recommended as the healthiest method to keep the database environments where such information is stored, especially the security patches up-to-date (Bertino and Sandhu, 2005; Morrison, 2003; Gupta and Agrawal, 2017).

Solutions for Threats to Data Privacy: The encryption method uses the encryption algorithm to store the data in the database in certain formats. Key structure is used during these encryptions. The data related to the determined key is encrypted and the data can be opened using the key again. In encryption methods, two different methods are adopted symmetric and asymmetric. In the symmetric method, a fixed key is used in encryption and decryption operations, while in the asymmetric structure, new passwords are used for each transaction both during encryption and decryption. Here, the management of these encryptions is of great importance. How often keys are changed, how they are managed, where they are stored, and where encryption is performed are vital issues. Storing keys at points with limited access provides a high level of security. At the same time, performing encryption operations on a separate server takes the encryption processing power from the main server. Generally supported encryption methods can be listed as DES, Triple DES, RC2, RC4, DESX, AES, and MD5 (Mousa et al., 2020; Bertino and Sandhu, 2005; Morrison, 2003; Gupta and Agrawal, 2017).

Many different security threats can pose a significant risk to the data of your organization and clients. The most common database threat is SQL injection, but attacks such as Denial of Service and malware are equally dangerous. Training your employees, using encryption, and managing user privileges are some of the best ways to protect your database from a cyberattack (Mallaboyev et al., 2022; Khan et al., 2022).

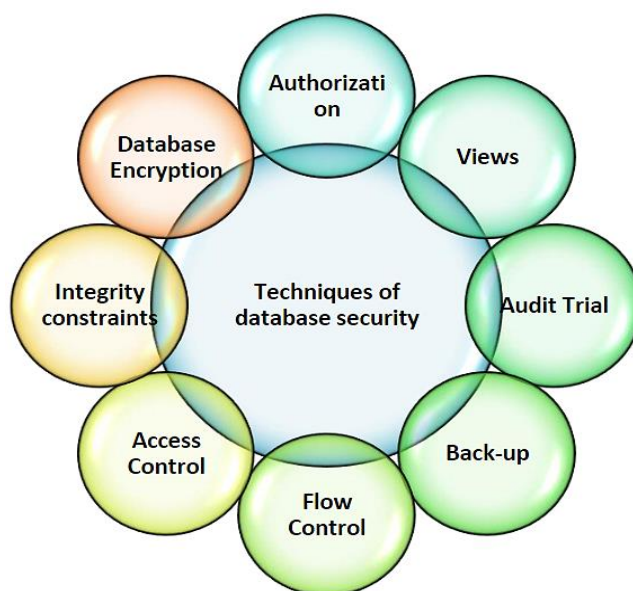


Figure 4. Various techniques for database security (Alruwaili, 2012).

Databases are almost always accessible over the network, a security threat to any component in or part of the network infrastructure is also a threat to the database, and an attack that affects a user's device or workstation can also threaten the database. Figure 4. shows some

techniques to protect the database but also considers the given below lists. Therefore, database security must reach far beyond database boundaries alone. Consider each of the following areas when evaluating database security in your environment to decide your team's top priorities (Avcı, 2021; Avcı, 2022):

- Physical security
- Management and network access controls
- End user account/device security
- Encryption
- Database software security
- Application/web server security
- Backup security
- Audit

4. CONCLUSIONS

As a result of the research, it was concluded that there are no current and sufficient resources for database questions and solution suggestions. The most important point here is that the necessary in-house and individual security training should be increased and the necessary academic studies should be done at the same time. To give the results of the suggestions and research given in this report, most of the problems arise from the lack of sufficient information infrastructure. Making the training packages to be prepared for at least simple measures as in the current report mandatory will be one of the biggest obstacles for all these attack structures. According to the research, database security problems are generally caused by not taking simple precautions. The security level can be taken to a higher level by taking many technical precautions in the shared information, but in addition, entry-level, partially user-based measures such as the strong definition of passwords, not using the same passwords, not using default user names, and keeping system structures up-to-date, which are much simpler methods, can be taken much easier. Taking the measures at the entry-level can bring the security level to a strong point. After the entry-level measures, the technical measures taken by a database administrator reveal a much stronger level of security. Database security requires layered security controls across your entire network environment, as well as establishing the right controls and policies for accessing the database itself. Now that you know of the most common database security threats, here are some helpful tips to protect your database:

- Train employees with risk mitigation techniques and best practices
- Manage user access rights, eliminate excessive privileges, and remove inactive users
- Access to any database vulnerabilities
- Block any malicious web requests
- Monitor all database access activity and usage patterns in real-time
- Archive your external data
- Mask database fields to hide any sensitive information
- Determining the attack surface of your network and database.
- Using a zero-trust system.
- Deleting inactive accounts and limiting privileges for standard users.
- Encrypting the database and all backups.
- Using masking to hide database fields that contain sensitive information.

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